ALICE LUNA

astroaliceluna@gmail.com ⋆ astroaliceluna.github.io

SUMMARY

Astrophysicist and STEM educator with 5+ years of teaching and community outreach. Experienced in Python, statistical analysis, and data visualization.

EDUCATION

University of Chicago Expected Spring 2027
Ph.D Astronomy & Astrophysics
Thesis – *A Deep Stellar Spectroscopic Analysis of Metallicity Distribution Functions of Ultra–Faint Dwarf Galaxies*
Advisor: Dr. Alexander Ji

University of Chicago December 2024
M.S. Astronomy & Astrophysics
Thesis – *The Metallicity Distribution Function of the Ultra–Faint Dwarf Galaxy Reticulum II*
Advisor: Dr. Alexander Ji

University of Texas at Austin May 2021
B.S. Astronomy & Physics
Relevant Courses – *Race, Science, Ethics and Justice in the Classroom; Knowing and Learning in Math and Science; Secondary Teacher Education Practicum I and II*

PUBLICATIONS

[1] **Alice M. Luna**, Alexander P. Ji, Anirudh Chiti, Joshua D. Simon, Daniel D. Kelson, Minsung Go, Guilherme Limberg, Ting S. Li, and Anna Frebel. "A Bimodal Metallicity Distribution Function in the Ultra-Faint Dwarf Galaxy Reticulum II". In: *The Open Journal of Astrophysics* 8 (Nov. 2025), p. 47696. DOI: 10.33232/001c.147696. arXiv: 2506.16462 [astro-ph.GA].

[2] Daisy Bissonette, Alexander P. Ji, Joshua D. Simon, Joss Bland-Hawthorn, Anirudh Chiti, Marla Geha, Ting S. Li, Anna Frebel, and **Alice M. Luna**. "The Metallicity Distribution of the Ultra-Faint Dwarf Galaxy Segue 1". In: *arXiv e-prints*, arXiv:2605.12810 (May 2026), arXiv:2605.12810. DOI: 10.48550/arXiv.2605.12810. arXiv: 2605.12810 [astro-ph.GA].

[3] William Cerny, Daisy Bissonette, Alexander P. Ji, Marla Geha, Anirudh Chiti, Simon E. T. Smith, Joshua D. Simon, Andrew B. Pace, Evan N. Kirby, Kim A. Venn, Ting S. Li, and **Alice M. Luna**. "No Observational Evidence for Dark Matter Nor a Large Metallicity Spread in the Extreme Milky Way Satellite Ursa Major III/UNIONS 1". In: 999.1, L8 (Mar. 2026), p. L8. DOI: 10.3847/2041-8213/ae29b8. arXiv: 2510.02431 [astro-ph.GA].

[4] Kaia R. Atzberger, Sam A. Usman, Alexander P. Ji, Lara R. Cullinane, Denis Erkal, Terese T. Hansen, Geraint F. Lewis, Ting S. Li, Guilherme Limberg, **Alice Luna**, Sarah L. Martell, Madeleine McKenzie, Andrew B. Pace, and Daniel B. Zucker. "Chemical Abundances in the Leiptr Stellar Stream: A Disrupted Ultra-faint Dwarf Galaxy?" In: *The Open Journal of Astrophysics* 8, 68 (June 2025), p. 68. DOI: 10.33232/001c.139013. arXiv: 2410.17312 [astro-ph.GA].

[5] Madeline Lucey, Keith Hawkins, Melissa Ness, Tyler Nelson, Victor P. Debattista, **Alice Luna**, Thomas Bensby, Kenneth C. Freeman, and Chiaki Kobayashi. "The COMBS Survey - III. The chemodynamical origins of metal-poor bulge stars". In: 509.1 (Jan. 2022), pp. 122–144. DOI: 10.1093/mnras/stab2878. arXiv: 2107.02793 [astro-ph.GA].

[6] Madeline Lucey, Keith Hawkins, Melissa Ness, Victor P. Debattista, **Alice Luna**, Martin Asplund, Thomas Bensby, Luca Casagrande, Sofia Feltzing, Kenneth C. Freeman, Chiaki Kobayashi, and Anna F. Marino. "The COMBS Survey - II. Distinguishing the metal-poor bulge from the halo interlopers". In: 501.4 (Mar. 2021), pp. 5981–5996. DOI: 10.1093/mnras/stab003. arXiv: 2009.03886 [astro-ph.GA].

PRESENTATIONS & POSTERS

SACNAS National Diversity in STEM Conference, **Presentation** — Oct 2026
Southside Astronomy Program, **Presentation** — Sep 2026
Astronomy Chalk Talks at UChicago, **Presentation** — Apr 2026
Near Field Cosmology at UChicago, **Presentation** — Nov 2025
Astronomy Chalk Talks at UChicago, **Presentation** — Apr 2025
Galactic Frontiers II – Dwarf Galaxies in the Local Volume & Beyond, **Poster** — Jun 2025
Magellan Science Meeting, **Presentation** — May 2025
Small Galaxies Cosmic Questions II, **Poster** — Aug 2024
Dwarf Galaxies, Star Clusters, and Streams in the LSST Era, **Poster** — Jul 2024
First Stars VII in NYC, **Poster** — May 2024
Near Field Cosmology at UChicago, **Presentation** — Feb 2024
Galactic Frontiers – Dwarf Galaxies in the Local Volume & Beyond, **Presentation** — Jul 2023
Frontiers in Nuclear Astrophysics, **Presentation** — May 2023

TEACHING EXPERIENCE

Instructor, *Clubes de Ciencia México, Universidad de Guanajuato* — Summer 2026
Designed and co-facilitated a week-long science course in Spanish "Desentrañando La Luz Estelar Con Código" for high school and college students on stellar spectroscopy and machine learning

Instructor, *SEED Program, SkAI Institute* — Summer 2026
Designed and facilitated an introductory lecture for undergraduate summer students on stellar astrophysics with a hands-on Python tutorial on classifying stellar spectra with machine learning

Instructor, *Astronomy Summer Program, UT Austin* — Spring 2026 – Summer 2026
Trained in evidence-based STEM pedagogy through a week-long intensive PDP at the Institute for Science and Engineer Educators, and co-designed and co-facilitated a 6-hour astronomy lab on astronomical imaging at the UT Austin Astronomy Summer Program.

Guest Lecturer, *Department of Astronomy & Astrophysics, UChicago*
ASTR 11901: Physics of Stars — Summer 2024
ASTR 11901: Physics of Stars — Summer 2023

Graduate Teaching Assistant, *Department of Astronomy & Astrophysics, UChicago*
ASTR 12720: Exoplanets — Spring 2022
ASTR 12710: Galaxies — Winter 2022
ASTR 12700: Stars — Fall 2021

Undergraduate Teaching Assistant, *Department of Astronomy, UT Austin*
AST 301: Introduction to Astronomy — Fall 2020, Spring 2021
AST 376: Observational Methods in Astronomy — Spring 2020
AST 103L: Astronomical Observations — Spring 2018, Fall 2018, Spring 2019, Fall 2019

Tutor, *Why Academy, UTeach Internship at UT Austin* — 2019 – 2021
Weekly online tutoring in math, science, and Spanish for middle and high school students from rural backgrounds with limited access to educational services

GRANTS, AWARDS & FELLOWSHIPS

Asterion Foundation Grant for Outreach Development	May 2026
Bridge Builder Award, UChicago	Spring 2025
NASA SCoPE SciAct Affiliate for SACNAS	Fall 2024
Physical Sciences Division Inclusive Climate Grant, UChicago	Summer 2024
NSF Graduate Research Fellowship (UChicago PSD News)	2022 – 2025
NSF Graduate Research Fellowship Honorable Mention	2021
Department of Astronomy Summer Research Fellowship, UT Austin	Summer 2020
Jewel Popham Raschke Endowed Presidential Scholarship, UT Austin	2019 – 2020
Walter E. Millett Endowed Scholarship in Physics, UT Austin	2019 – 2020
TIDES Advancement Summer Research Fellowship, UT Austin	Summer 2019
John W. Cox Endowment for Advanced Studies in Astronomy, UT Austin	Summer 2019
College of Natural Sciences Second Year Excellence Award, UT Austin	2019
Jane Sanford Beasley Scholarship for UTeach, UT Austin	2018
University Honors, UT Austin	2017 – 2020

COMMUNITY ENGAGEMENT & OUTREACH

Southside Astronomy Program, Co–Founder	Spring 2025 – Present
Inclusion, Diversity, and Equity in Astronomy, Organizer	Spring 2025 – Present
Society of Hispanic Professional Engineers, Organizer	Fall 2024 – Spring 2025
UChicago Astronomy Community Engagement Group, Organizer	Summer 2022 – Present
Board for Student Advocacy, Member	Spring 2020 – Spring 2021
Girls Invent Program, Mentor	Fall 2023
Graduate Undergraduate Mentorship in Astronomy, Mentor	Fall 2019 – Spring 2021
Undergraduate Gender Minorities in Physics, Vice President	Spring 2019 – Spring 2020
Astronomy Students Association, Treasurer	Spring 2018 – Spring 2019

SKILLS

Languages: English (Native), Spanish (Intermediate), American Sign Language (Basic)
Python Programming, LaTeX

REFERENCES

Dr. Alexander Ji, Assistant Professor
University of Chicago, Department of Astronomy & Astrophysics
Email: alexji@uchicago.edu
Relationship: Ph.D. Advisor

Dr. Juliet Crowell, Director of Public Education and Outreach
University of Chicago, Department of Astronomy & Astrophysics
Email: crowellj@uchicago.edu
Relationship: Outreach Collaborator